

CineStream SQL Project Report

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Part A: Core CRUD Operations

Objective:

To perform Create, Read, Update, and Delete operations on the CineStream database.

1. Create Operation (Insert)

Query:

```
INSERT INTO Movie (Title, Genre, Release_year, Rental_rate)
VALUES ('Interstellar', 'Sci-Fi', 2014, 4.99);
```

Result:

A new movie record was successfully added.

2. Read Operation (Select)

Query:

```
SELECT * FROM Users
WHERE Join_date > '2025-01-01'
AND Status = 'Active';
```

Result:

The query returned an empty set because no users matched the conditions.

3. Update Operation

Query:

```
UPDATE Users
SET Status = 'Active'
WHERE Username = 'Selena Gomez';
```

Result:

The status of Selena Gomez was successfully updated to Active.

4. Delete Operation

Query:

```
DELETE FROM Users
WHERE Status = 'Inactive'
AND User_id NOT IN (
    SELECT User_id FROM WatchLogg
);
```

Result:

Inactive users without watch history can be removed from the database.

Conclusion:

Part A successfully demonstrated Create, Read, Update, and Delete operations.

```
mysql> desc Movieess;
+-----+-----+-----+-----+-----+-----+
| Field      | Type      | Null | Key | Default | Extra      |
+-----+-----+-----+-----+-----+-----+
| movie_id   | int       | NO   | PRI | NULL    | auto_increment |
| Title      | varchar(100) | YES  |     | NULL    |              |
| Genre      | varchar(60)  | NO   |     | NULL    |              |
| Release_year | int       | YES  |     | NULL    |              |
| Rental_rate | decimal(10,0) | YES  |     | NULL    |              |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)

mysql> desc WatchLogg;
+-----+-----+-----+-----+-----+-----+
| Field      | Type      | Null | Key | Default | Extra      |
+-----+-----+-----+-----+-----+-----+
| log_id     | int       | NO   | PRI | NULL    | auto_increment |
| user_id    | int       | YES  |     | NULL    |              |
| movie_id   | int       | YES  |     | NULL    |              |
| watch_date | datetime  | YES  |     | NULL    |              |
| duration_minutes | int       | YES  |     | NULL    |              |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)

mysql> desc Users;
+-----+-----+-----+-----+-----+-----+
| Field      | Type      | Null | Key | Default | Extra      |
+-----+-----+-----+-----+-----+-----+
| User_id    | int       | NO   | PRI | NULL    | auto_increment |
| Username   | varchar(70) | NO   |     | NULL    |              |
| Email      | varchar(60) | NO   | UNI | NULL    |              |
| Join_date  | date      | YES  |     | NULL    |              |
| Status     | varchar(80) | YES  |     | NULL    |              |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM Users
  -> WHERE Username='Selena Gomez';
+-----+-----+-----+-----+-----+
| User_id | Username | Email | Join_date | Status |
+-----+-----+-----+-----+-----+
| 10 | Selena Gomez | gomezz@gmail.com | 2024-06-19 | Active |
+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)

mysql> DELETE FROM Users
  -> WHERE Status='Inactive'
  -> AND User_id NOT IN (
  -> SELECT user_id
  -> FROM WatchLogg
  -> );
Query OK, 0 rows affected (0.02 sec)

mysql> SELECT * FROM Users;
+-----+-----+-----+-----+-----+
| User_id | Username | Email | Join_date | Status |
+-----+-----+-----+-----+-----+
| 1 | Conan Gray | conan@gmail.com | 2006-07-24 | Active |
| 2 | Sabrina Carpenter | sabrina@gmail.com | 2022-05-04 | Inactive |
| 3 | Lana del Rey | lanadidi@gmail.com | 2019-06-08 | Active |
| 4 | Nicki Minaj | nicki@gmail.com | 2012-11-15 | Active |
| 5 | Taylor Swift | swift@gmail.com | 2010-06-12 | Inactive |
| 6 | Ariana Grande | ariana@gmail.com | 2013-05-22 | Active |
| 7 | Drake | Drake@gmail.com | 2015-06-17 | Inactive |
| 8 | Gracie Abrams | grace@gmail.com | 2018-06-09 | Active |
| 9 | Kevv | Kevv@gmail.com | 2016-09-24 | Inactive |
| 10 | Selena Gomez | gomezz@gmail.com | 2024-06-19 | Active |
+-----+-----+-----+-----+-----+
10 rows in set (0.00 sec)

mysql> SELECT
  -> m.Genre,
```

Part B: Analytical JOIN Operations

Genre Popularity Report

Objective:
To identify the most popular movie genres based on user viewing activity.

SQL Query:
SELECT m.Genre AS Genre_Name,
COUNT(w.log_id) AS Total_Watch_Count
FROM Movieess m
INNER JOIN WatchLogg w
ON m.movie_id = w.movie_id
GROUP BY m.Genre
ORDER BY Total_Watch_Count DESC;

Output Summary:
Action - 2
Sci-Fi - 2
Romance - 1
Adult - 1
Horror - 1
Romcom - 1
Romance - 1
Tragedy - 1

Conclusion:

Action and Sci-Fi are the most popular genres. CineStream should consider licensing more content in these categories to improve engagement and growth.

```
10 rows in set (0.00 sec)

mysql> SELECT
  -> m.Genre,
  -> COUNT(w.log_id) AS Total_Watch_Count
  -> FROM Movies$ m
  -> INNER JOIN WatchLog$ w
  -> ON m.movie_id = w.movie_id
  -> GROUP BY m.Genre
  -> ORDER BY Total_Watch_Count DESC;
+-----+-----+
| Genre | Total_Watch_Count |
+-----+-----+
| Sci-Fi | 2 |
| Action | 2 |
| Romance | 1 |
| Adult | 1 |
| horror | 1 |
| Romcom | 1 |
| Rommance | 1 |
| Tragedy | 1 |
+-----+-----+
8 rows in set (0.02 sec)

mysql> desc Movies$;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| movie_id | int | NO | PRI | NULL | auto_increment |
| Title | varchar(100) | YES | | NULL | |
| Genre | varchar(60) | NO | | NULL | |
| Release_year | int | YES | | NULL | |
| Rental_rate | decimal(10,0) | YES | | NULL | |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)
```